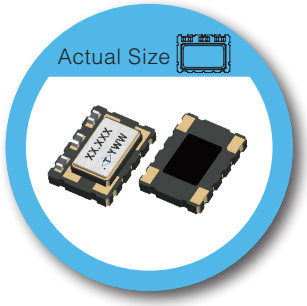
TAITIEN
Our Performance, Your Reputation

TT Type < for Stratum 3 >

7.0 x 5.0 mm SMD Voltage Controlled Temperature Compensated Crystal Oscillator



RoHS Compliant

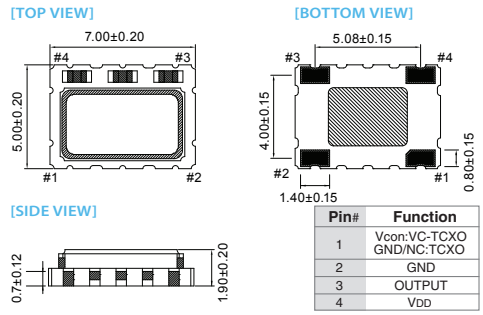
FEATURE

- Typical 7.0 x 5.0 x 1.9 mm ceramic SMD package.
- Stratum 3 (Overall ±4.6ppm including 20 years aging.)
- CMOS and Clipped Sine wave (without DC-cut capacitor) output optional.

TYPICAL APPLICATION

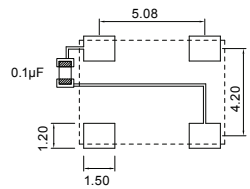
- Stratum 3
- Femtocell, Base Stations

DIMENSION (mm)



Pin#	Function
1	Vcon:VC-TCXO GND/NC:TCXO
2	GND
3	OUTPUT
4	VDD

SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1μF as close to the part as possible between Vdd and GND pads.

ELECTRICAL SPECIFICATION

Parameter	2.5V		3.3V		Unit
	Min.	Max.	Min.	Max.	
Supply Voltage Variation (VDD)	VDD-5%	VDD+5%	VDD-5%	VDD+5%	V
Frequency Range	10	52	10	52	MHz
Standard Frequency	10, 12.8, 16.384, 19.2, 20, 25, 26				
Operating Temp. Range	-20 ~ 70 -40 ~ 85				°C
Frequency Stability (Overall, 20 Years)*	–	±4.6	–	±4.6	ppm
Frequency Stability Vs Temp. Range (Ref. to (FMAX+Fmin)/2)	–	±0.14 (-20~+70°C) ±0.28 (-40~+85°C)	–	±0.14 (-20~+70°C) ±0.28 (-40~+85°C)	ppm
Holdover Stability +	–	±0.32	–	±0.32	ppm
Supply Current (CMOS output)	–	8	–	8	mA
Supply Current (Clipped Sine Wave)	–	5	–	5	
Output Level (CMOS) Output High (Logic "1")	90%VDD	–	90%VDD	–	V
Output Low (Logic "0")	–	10%VDD	–	10%VDD	
Duty	45	55	45	55	%
Output Level (Clipped Sine Wave)	0.8	–	0.8	–	Vp-p
Load (CMOS)	15pF		15pF		
Load (Clipped Sine Wave)	10 KΩ // 10pF		10 KΩ // 10pF		
Control Voltage Range (VCTCXO)	0.5	2.5	0.5	2.5	V
Pulling Range (VCTCXO)	±5.0	–	±5.0	–	ppm
Vc Input Impedance (VCTCXO)	100	–	100	–	kΩ
Phase Noise @ 10 MHz 100 Hz	-130				dBc/Hz
1 kHz	-145				
10 kHz	-154				
Start time	–	10	–	10	mSec
Storage Temp. Range	-55	125	-55	125	°C

Standard frequencies are frequencies which the crystal has been designed and does not imply a stock position.

* Including calibration @ 25°C, supply voltage VDD±5% , load ±10%, reflow soldering, 20 years aging and frequency stability over temperature.

+ Including 24hours aging , supply voltage VDD±5% and frequency stability over temperature.

Note: not all combination of options are available. Other specifications may be available upon request.